

Fundamental Concepts in Inference

Point Estimation — Confidence Sets

Md Irtiaz Kabir

Point Estimation — What Is It?

Point estimation: provide a single “best guess” of some quantity of interest.

The quantity could be:

- A parameter θ in a parametric model
- A CDF F , a density f , a regression function r
- A prediction for a future observation Y

Convention

We denote a point estimate of θ by $\hat{\theta}$ or $\hat{\theta}_n$.

θ is *fixed and unknown*. $\hat{\theta}$ *depends on the data*, so it is a **random variable**.

The subscript n denotes the *sample size*

Some Notation

If $\mathfrak{F} = \{f(x; \theta) : \theta \in \Theta\}$ is a **parametric model**, we write

$$\mathbb{P}_\theta(X \in A) = \int_A f(x; \theta) dx \quad \text{and} \quad \mathbb{E}_\theta(r(X)) = \int r(x) f(x; \theta) dx.$$

Important

The subscript θ indicates that the probability or expectation is *with respect to* $f(x; \theta)$. It does **not** mean we are averaging over θ .

Similarly, we write $\mathbb{V}_\theta$ for the variance.

Formal Setup

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$.

A **point estimator** $\hat{\theta}_n$ of θ is any function of the data:

$$\hat{\theta}_n = g(X_1, \dots, X_n).$$

Two fundamental properties:

Bias

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}_{\theta}(\hat{\theta}_n) - \theta$$

$\hat{\theta}_n$ is **unbiased** if $\mathbb{E}(\hat{\theta}_n) = \theta$.

Standard Error

The **sampling distribution** of $\hat{\theta}_n$ has standard deviation

$$\text{se} = \text{se}(\hat{\theta}_n) = \sqrt{\mathbb{V}(\hat{\theta}_n)}.$$

Estimated version denoted $\hat{\text{se}}$.

Intuition — Bias and Standard Error

Think of $\hat{\theta}_n$ as a **dart-throwing** process: each dataset gives one throw at the target θ .

Bias = off-target

How far the *average* throw is from the bullseye.

Low bias: throws are centered on θ , even if scattered.

SE = spread

How much the throws *scatter* from one dataset to the next.

Low SE: throws land close together, even if off-center.

Key point

Bias and SE are **different notions of error**: a biased-but-precise estimator is consistently wrong; an unbiased-but-noisy estimator is right *on average* but unreliable on any single dataset.

Some More Notations: Convergence in Probability

Definition

X_n **converges to X in probability**, written $X_n \xrightarrow{P} X$, if for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$$

as $n \rightarrow \infty$.

Intuition: as n grows, it becomes increasingly unlikely that X_n is far from X .

Definition

X_n **converges to X in quadratic mean**, written $X_n \xrightarrow{qm} X$,

$$\mathbb{E}(X_n - X)^2 \rightarrow 0$$

as $n \rightarrow \infty$.

Intuition: as n grows, the “average squared difference” between X_n and X starts to become negligible.

Implications Between the Two Convergence

$X_n \xrightarrow{qm} X$ implies that $X_n \xrightarrow{P} X$

Proof. Suppose $X_n \xrightarrow{qm} X$. Fix $\epsilon > 0$. Then, using Markov's inequality,

$$\mathbb{P}(|X_n - X| > \epsilon) = \mathbb{P}(|X_n - X|^2 > \epsilon^2) \leq \frac{\mathbb{E}|X_n - X|^2}{\epsilon^2} \rightarrow 0.$$

Definition — Consistency

A point estimator $\hat{\theta}_n$ is **consistent** if

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

Unbiasedness used to receive much attention but is now considered less important — many useful estimators are biased.

Consistency is a more natural requirement: the estimator should converge to the truth as we collect more data.

Example — Bernoulli Estimator (Bias)

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$ and $\hat{p}_n = n^{-1} \sum_{i=1}^n X_i$. For a single X_i : $\mathbb{E}(X_i) = p$, $\mathbb{V}(X_i) = p(1 - p)$.

Computing $\mathbb{E}(\hat{p}_n)$

$$\begin{aligned}\mathbb{E}(\hat{p}_n) &= \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) && \text{(linearity of expectation)} \\ &= \frac{1}{n} \sum_{i=1}^n p = \frac{1}{n}(np) = p.\end{aligned}$$

$\text{bias}(\hat{p}_n) = \mathbb{E}(\hat{p}_n) - p = p - p = 0 \Rightarrow \hat{p}_n$ is **unbiased**.

Example — Bernoulli Estimator (Variance & SE)

Since $X_1, \dots, X_n$ are **independent**, $\mathbb{V}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \mathbb{V}(X_i)$.

Computing $\mathbb{V}(\hat{p}_n)$

$$\mathbb{V}(\hat{p}_n) = \mathbb{V}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \mathbb{V}\left(\sum_{i=1}^n X_i\right) \quad (\text{const. factor, squared})$$

$$= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(X_i) \quad (\text{independence})$$

$$= \frac{1}{n^2} (n p(1-p)) = \frac{p(1-p)}{n}.$$

- $\text{se} = \sqrt{\mathbb{V}(\hat{p}_n)} = \sqrt{p(1-p)/n}.$
- p unknown $\Rightarrow$ plug in $\hat{p}_n$: Estimated standard error $\hat{\text{se}} = \sqrt{\hat{p}_n(1-\hat{p}_n)/n}.$

Mean Squared Error (MSE)

The quality of an estimator is often assessed by

$$\text{MSE} = \mathbb{E}_{\theta}(\hat{\theta}_n - \theta)^2.$$

Important

$\mathbb{E}_{\theta}(\cdot)$ is expectation w.r.t. the data distribution $f(x_1, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta)$. It does *not* average over a distribution for θ .

Theorem — Bias–Variance Decomposition

$$\text{MSE} = \text{bias}^2(\hat{\theta}_n) + \mathbb{V}_{\theta}(\hat{\theta}_n).$$

Proof — Bias–Variance Decomposition

Let $\bar{\theta}_n = \mathbb{E}_\theta(\hat{\theta}_n)$, a *constant* (not random) for fixed θ . Add and subtract $\bar{\theta}_n$ inside the square:

$$\begin{aligned}\mathbb{E}_\theta(\hat{\theta}_n - \theta)^2 &= \mathbb{E}_\theta[(\hat{\theta}_n - \bar{\theta}_n) + (\bar{\theta}_n - \theta)]^2 \\&= \mathbb{E}_\theta[(\hat{\theta}_n - \bar{\theta}_n)^2 + 2(\hat{\theta}_n - \bar{\theta}_n)(\bar{\theta}_n - \theta) + (\bar{\theta}_n - \theta)^2] \\&= \mathbb{E}_\theta(\hat{\theta}_n - \bar{\theta}_n)^2 + 2(\bar{\theta}_n - \theta) \mathbb{E}_\theta(\hat{\theta}_n - \bar{\theta}_n) + (\bar{\theta}_n - \theta)^2 \\&= \mathbb{E}_\theta(\hat{\theta}_n - \bar{\theta}_n)^2 + 2(\bar{\theta}_n - \theta) \underbrace{[\mathbb{E}_\theta(\hat{\theta}_n) - \bar{\theta}_n]}_{=0} + (\bar{\theta}_n - \theta)^2 \\&= \underbrace{\mathbb{E}_\theta(\hat{\theta}_n - \bar{\theta}_n)^2}_{=\mathbb{V}_\theta(\hat{\theta}_n)} + \underbrace{(\bar{\theta}_n - \theta)^2}_{=\text{bias}^2(\hat{\theta}_n)} \\&= \text{bias}^2(\hat{\theta}_n) + \mathbb{V}_\theta(\hat{\theta}_n). \quad \square\end{aligned}$$

Theorem

If $\text{bias} \rightarrow 0$ and $\text{se} \rightarrow 0$ as $n \rightarrow \infty$, then $\hat{\theta}_n$ is consistent:

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

Proof sketch. By the bias–variance decomposition, $\text{MSE} \rightarrow 0$, so $\hat{\theta}_n \xrightarrow{\text{qm}} \theta$.
Convergence in quadratic mean implies convergence in probability. $\square$

Bernoulli revisited

$\text{bias}(\hat{p}_n) = 0$ and $\text{se} = \sqrt{p(1-p)/n} \rightarrow 0$, so $\hat{p}_n \xrightarrow{P} p$. $\checkmark$

Relation Between Unbiased and Consistent

- Does unbiased $\implies$ consistent?
- Does consistent $\implies$ unbiased?

Actually, none of these are true. Can you provide some example where -

- Some estimate is unbiased, yet inconsistent?
- Some estimate is consistent, yet biased?

Unbiased Yet Inconsistent

Consider this scenario -

- Samples: $X_1, X_2, \dots, X_n$
- Estimate $\hat{\mu} : X_1$

Basically, we always return the first value from our sample.

$$\text{Bias} = \mathbb{E}_{\theta}(\hat{\mu}) - \mu$$

$$\implies \mathbb{E}_{\theta}(X_1) - \mu$$

$$\implies \mu - \mu = 0$$

$\therefore$ Unbiased

But do you think, always returning the first element will perform “better” as more and more data is collected?

Definition

A $1 - \alpha$ **confidence interval** for θ is an interval $C_n = (a, b)$, where $a = a(X_1, \dots, X_n)$ and $b = b(X_1, \dots, X_n)$, such that

$$\mathbb{P}_\theta(\theta \in C_n) \geq 1 - \alpha \quad \text{for all } \theta \in \Theta.$$

- $1 - \alpha$ is called the **coverage** of the interval.
- Common choice: $\alpha = 0.05$ (95% confidence interval).
- For vector θ : use a **confidence set** (sphere, ellipse, ...).

Key Point

C_n is **random**; θ is **fixed**.

Interpreting Confidence Intervals

Wrong interpretation: “There is a 95% probability that θ lies in C_n .” (θ is not random.)

Correct but impractical: If you repeat the experiment many times, the interval will contain θ in 95% of repetitions.

Better interpretation (Wasserman)

Suppose you construct a 95% CI for an *unrelated* parameter each day. Over many days, **95% of those intervals will contain the true parameter**. No need to imagine repeating the same experiment.

Example — Opinion Polls

Newspapers report: *“83% of people favour arming pilots with guns, accurate to within 4 points 95% of the time.”*

Translation: 83 ± 4 is a **95% confidence interval** for the true proportion p .

If you construct such an interval every day (for different poll questions), 95% of your intervals will contain the true proportion — even though you estimate a different quantity each time.

Example — CI Is Not a Probability Statement

(Berger & Wolpert, 1984.) Let θ be fixed and X_1, X_2 be independent with $\mathbb{P}(X_i = \pm 1) = 1/2$. Observe $Y_i = \theta + X_i$.

Define:

$$C = \begin{cases} \{Y_1 - 1\} & \text{if } Y_1 = Y_2, \\ \{(Y_1 + Y_2)/2\} & \text{if } Y_1 \neq Y_2. \end{cases}$$

One can verify: $\mathbb{P}_\theta(\theta \in C) = 3/4$ for all θ .

Observed: $Y_1 = 15, Y_2 = 17 \Rightarrow C = \{16\}$, and we are *certain* $\theta = 16$.

Yet $\{16\}$ is still only a 75% CI. A confidence level is a **pre-experiment** guarantee, not a post-experiment probability.

Verifying $\mathbb{P}_\theta(\theta \in C) = 3/4$

(X_1, X_2) has 4 equally likely outcomes (prob. $1/4$ each), since X_1, X_2 are independent and each is ± 1 with probability $1/2$. Recall $Y_i = \theta + X_i$.

(X_1, X_2)	(Y_1, Y_2)	Rule used	C	$\theta \in C?$
$(1, 1)$	$(\theta+1, \theta+1)$	$Y_1=Y_2 \Rightarrow \{Y_1-1\}$	$\{\theta\}$	Yes ✓
$(1, -1)$	$(\theta+1, \theta-1)$	$Y_1 \neq Y_2 \Rightarrow \{(Y_1+Y_2)/2\}$	$\{\theta\}$	Yes ✓
$(-1, 1)$	$(\theta-1, \theta+1)$	$Y_1 \neq Y_2 \Rightarrow \{(Y_1+Y_2)/2\}$	$\{\theta\}$	Yes ✓
$(-1, -1)$	$(\theta-1, \theta-1)$	$Y_1=Y_2 \Rightarrow \{Y_1-1\}$	$\{\theta-2\}$	No ✕

Why the last row fails

When $X_1 = X_2 = -1$, both observations undershoot θ by 1, but the rule $C = \{Y_1 - 1\}$ was calibrated for the $X_1 = X_2 = 1$ case (where it shifts *up*) — so here it shifts the wrong way and misses θ by 2.

$\theta \in C$ in 3 of the 4 equally likely cases $\Rightarrow \mathbb{P}_\theta(\theta \in C) = 3/4$.

Normal-Based Confidence Interval

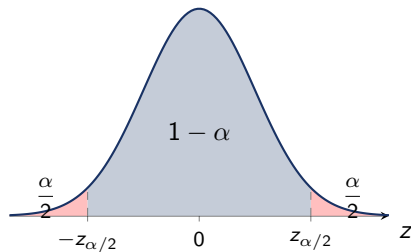
Theorem

Suppose $\hat{\theta}_n \approx N(\theta, \widehat{se}^2)$. Let $z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2)$, so $\mathbb{P}(Z > z_{\alpha/2}) = \alpha/2$ for $Z \sim N(0, 1)$. Define

$$C_n = (\hat{\theta}_n - z_{\alpha/2} \widehat{se}, \hat{\theta}_n + z_{\alpha/2} \widehat{se}).$$

Then $\mathbb{P}_\theta(\theta \in C_n) \rightarrow 1 - \alpha$.

For 95% intervals: $\alpha = 0.05$,
 $z_{\alpha/2} = 1.96 \approx 2$, giving the familiar $\hat{\theta}_n \pm 2 \widehat{se}$.



Standard normal density $\varphi(z)$: middle area is the coverage, tail areas are the error budget.

Let $Z_n = (\hat{\theta}_n - \theta)/\widehat{\text{se}}$. By assumption $Z_n \rightsquigarrow N(0, 1)$.

$$\begin{aligned}\mathbb{P}_\theta(\theta \in C_n) &= \mathbb{P}_\theta\left(\hat{\theta}_n - z_{\alpha/2}\widehat{\text{se}} < \theta < \hat{\theta}_n + z_{\alpha/2}\widehat{\text{se}}\right) \\ &= \mathbb{P}_\theta\left(-z_{\alpha/2} < \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} < z_{\alpha/2}\right) \\ &\rightarrow \mathbb{P}(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha. \quad \square\end{aligned}$$

Example — Bernoulli Normal-Based CI

$$X_1, \dots, X_n \sim \text{Bernoulli}(p), \quad \hat{p}_n = n^{-1} \sum_{i=1}^n X_i.$$

- $\text{se} = \sqrt{p(1-p)/n}$, $\hat{\text{se}} = \sqrt{\hat{p}_n(1-\hat{p}_n)/n}$.
- By the CLT: $\hat{p}_n \approx N(p, \hat{\text{se}}^2)$.

Approximate $1 - \alpha$ confidence interval:

$$\hat{p}_n \pm z_{\alpha/2} \hat{\text{se}} = \hat{p}_n \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}}.$$